

Exploring Generalizations of the Collatz Algorithm: An Amateur Investigation

Mohamed Amine Belachhab

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Abstract

This paper presents an amateur exploration of potential generalizations of the Collatz conjecture algorithm. While the author acknowledges this is preliminary work by a high school student with limited resources, the investigation attempts to understand the structural properties that might allow the Collatz algorithm to terminate at 1 for all positive integers. We propose a generalized framework and analyze specific cases, though we note that the mathematical rigor requires further development.

1 Introduction

The Collatz conjecture, one of mathematics' most famous unsolved problems, concerns the behavior of the following algorithm applied to any positive integer n :

- If n is even, divide it by 2
- If n is odd, multiply by 3 and add 1

The conjecture states that this process will eventually reach 1 for any starting positive integer.

This paper attempts to generalize this algorithm and understand what conditions might guarantee convergence to 1. We emphasize that this is exploratory work that would benefit from more rigorous mathematical treatment.

2 Proposed Generalization

We consider the following generalized algorithm for positive integers n , with parameters x , k , and α :

Algorithm G(x, k, α):

- If x divides n , then replace n with n/x
- Otherwise, replace n with $kn + (n \pm \alpha)/x$

Where:

- x, k are positive integers
- α is chosen as the smallest positive integer such that $n \pm \alpha$ is divisible by x
- All operations must preserve membership in the positive integers

Objective: Determine values of x , k , and α such that this algorithm converges to 1 for all positive integer inputs.

3 Analysis of Convergence Conditions

3.1 Necessary Conditions for Reaching 1

For the algorithm to reach 1 from any starting value, we require that at some point in the sequence, we can apply the division step to get 1. This means we need:

- Either $n = x$ (so $n/x = 1$)
- Or $n = x^2$ (so $n/x = x$, then $x/x = 1$)

3.2 Case Analysis

Case 1: Algorithm produces x^2 in the sequence

If $kn + (n \pm \alpha)/x = x^2$, then after division by x , we get x .

This gives us:

$$kn + \frac{n \pm \alpha}{x} = x^2$$

Multiplying by x :

$$kxn + n \pm \alpha = x^3$$

Factoring:

$$n(kx + 1) \pm \alpha = x^3$$

For this to have integer solutions for general n , we need specific relationships between k , x , and α .

Case 2: Algorithm produces x in the sequence

If $kn + (n \pm \alpha)/x = x$, then after division by x , we get 1.

This gives us:

$$kxn + n \pm \alpha = x^2$$

Factoring:

$$n(kx + 1) \pm \alpha = x^2$$

3.3 Special Cases: $x = 2$ and $x = 3$

When $x = 2$ (related to the original Collatz conjecture):

- The division step is $n \rightarrow n/2$
- The modification step becomes $kn + (n \pm \alpha)/2$

When $x = 3$:

- The division step is $n \rightarrow n/3$
- The modification step becomes $kn + (n \pm \alpha)/3$

4 Limitations and Open Questions

4.1 Mathematical Rigor Issues

This analysis has several limitations:

1. The convergence arguments are not rigorous
2. The case analysis doesn't prove the algorithm works for all inputs
3. The relationship between parameters needs more careful treatment

4.2 Computational Considerations

The proposed algorithm modification:

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If = n mod x, then:  
- If x/2, set = -  
- If x/2, set = x -
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This attempts to minimize the adjustment needed, but convergence is not guaranteed.

5 Conclusion and Future Work

This preliminary investigation suggests that generalizing the Collatz algorithm faces significant challenges. The original Collatz conjecture's structure ($x = 2$, multiply by 3 and add 1 for odd numbers) appears to have special properties that are difficult to replicate in other parameter combinations.

Future research directions:

1. Develop more rigorous convergence proofs
2. Investigate computational behavior of proposed generalizations
3. Study cycle detection and divergence conditions
4. Explore connections to number theory and dynamical systems

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References

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